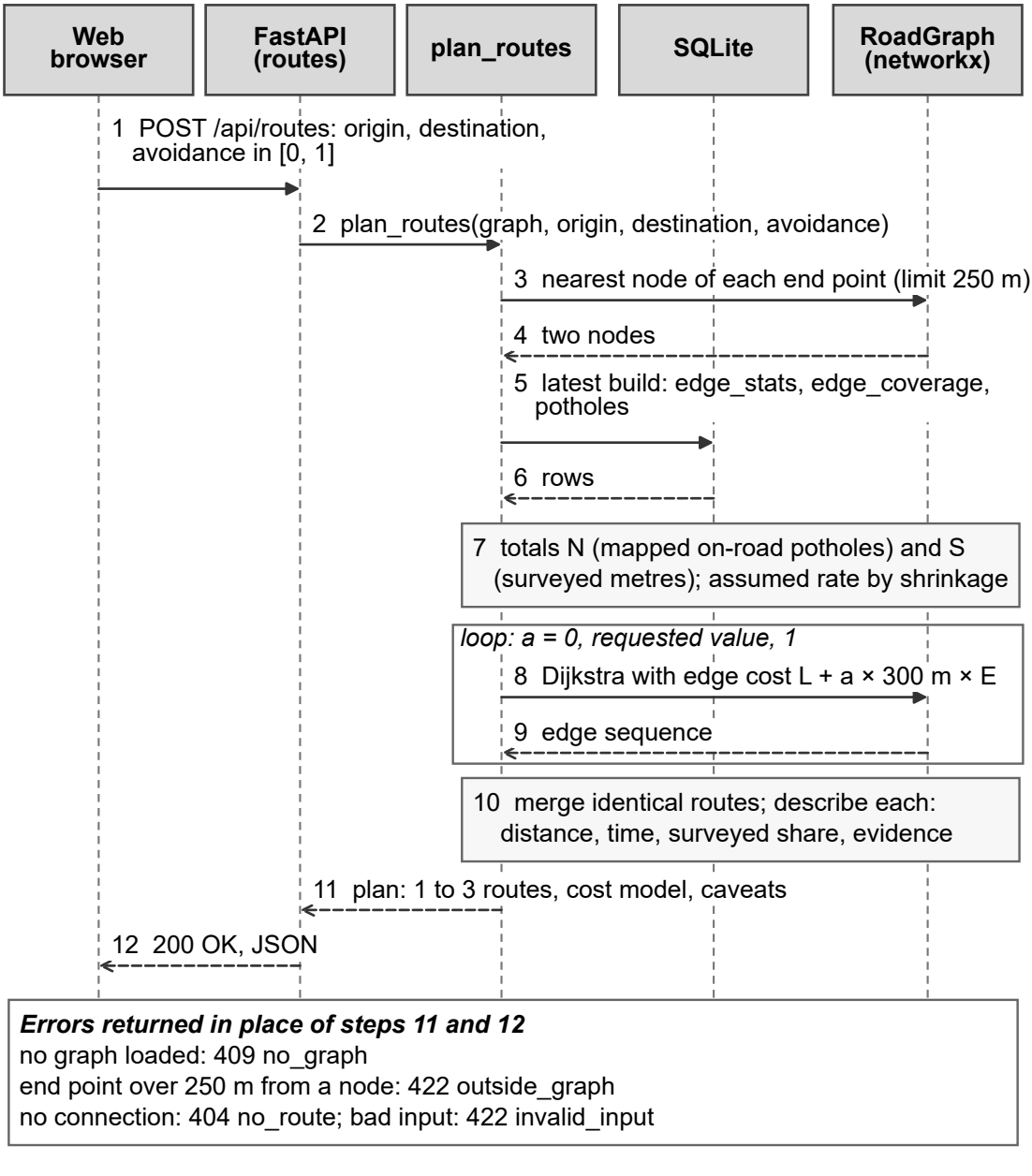

Web browser
FastAPI (routes)
plan_routes
SQLite
RoadGraph (networkx)
1 POST /api/routes: origin, destination, avoidance in [0, 1]
2 plan_routes(graph, origin, destination, avoidance)
3 nearest node of each end point (limit 250 m)
4 two nodes
5 latest build: edge_stats, edge_coverage, potholes
6 rows
7 totals N (mapped on-road potholes) and S (surveyed metres); assumed rate by shrinkage
loop: a = 0, requested value, 1
8 Dijkstra with edge cost L + a × 300 m × E
9 edge sequence
10 merge identical routes; describe each: distance, time, surveyed share, evidence
11 plan: 1 to 3 routes, cost model, caveats
12 200 OK, JSON
Errors returned in place of steps 11 and 12
no graph loaded: 409 no_graph
end point over 250 m from a node: 422 outside_graph
no connection: 404 no_route; bad input: 422 invalid_input